

EE 476

Multistage Diff-Amplifier

Final Report

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Abstract

This project aims to design a Multistage Differential Amplifier. The requirement of the project is to achieve 10dB, cutoff frequency, or -3 dB frequency, should be greater than 200MHz, CMRR should be at least 100, and 1% linearity needs to be achieved. To achieve these requirements, two stage OTA is used. The gain is $\sim 24\text{dB}$, and the -3 dB frequency is 240 MHz. The common mode gain is $\sim -13\text{dB}$, thus the CMRR is over 100. 1% linearity is also achieved after doing FFT analysis over a few frequencies. Two stage OTA is the design that matches the requirements.

Introduction

Operational transconductance amplifier, or OTA is an operational amplifier that has high gain, high input impedance, and high output impedance. OTA's output is the current source. Two stage amplifier allows for higher gain without reducing the output swing. The first stage is a differential amplifier and the second stage is typically configured as a simple common source amplifier. Two stage OTA design in the EE 476 class slide is in Figure 1.

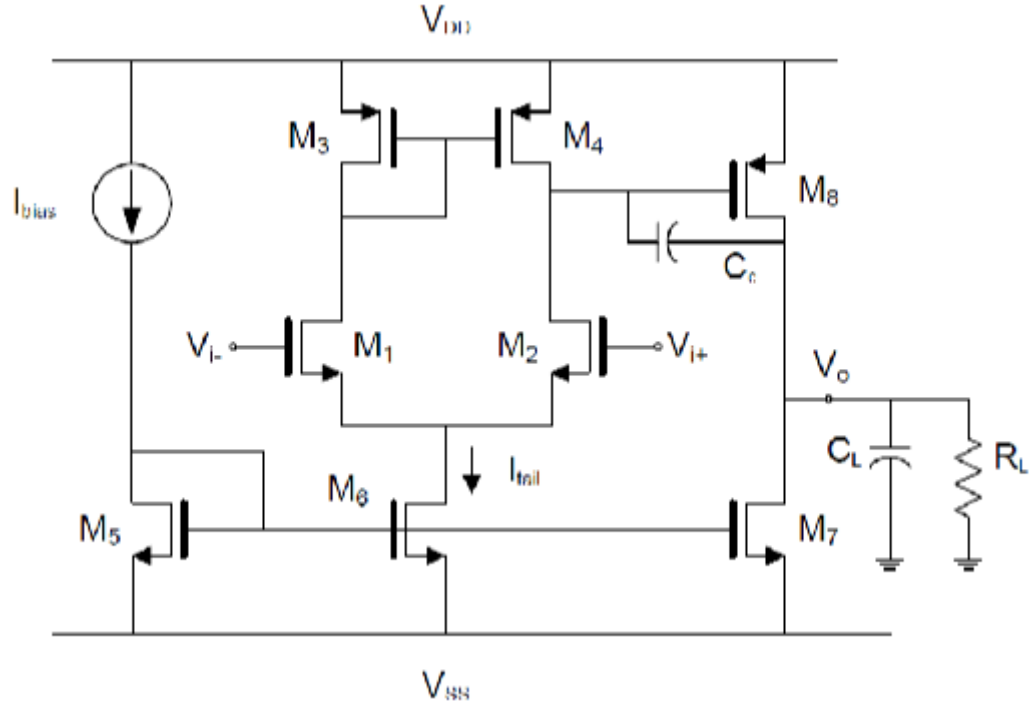


Figure 1. Two Stage Miller OTA

The gain formula is: $A_v = A_{V1}A_{V2} = \left(-\frac{g_{m2}}{g_{o2}+g_{o4}}\right)\left(-\frac{g_{m8}}{g_{o8}+g_{o7}}\right) = \frac{g_{m2}g_{m8}}{(g_{o2}+g_{o4})(g_{o8}+g_{o7})}$

The cutoff frequency is where the gain decreases by -3 dB from the DC gain. The dB gain formula is: $dB = 20 \log A_v$. Common Mode Rejection Ratio, or CMRR is the ratio between differential-mode gain to common-mode gain. 1% linearity is calculated with

following formula: $linearity\ percentage = \frac{\sqrt{V_2^2 + V_3^2 + V_4^2}}{V_1} * 100\%$.

Design Considerations

My primary design consideration was to create a properly biased differential pair that I would later add a second stage to create an OTA. I established I-Tail using an R-bias resistor of value 10K which resulted in a gate voltage of -411mV to M13 (tail transistor). The sizing between M13 and M9 was done so in a way that ~1mA was being used to bias the differential pair. The sizing between the NMOS and the PMOS was done in a manner to ensure that we

Figure 1: OTA Circuit Schematic used as a reference for this design [1]

met the 10dB gain spec without losing out on bandwidth. The second stage added more gain to the output and slightly reduced the bandwidth. The sizing between M4 and M7 was done in a way to ensure that neither of the two fall out of saturation and enough current is being supplied as bias. The load capacitance was a design requirement. Reference Figure 2 for these considerations.

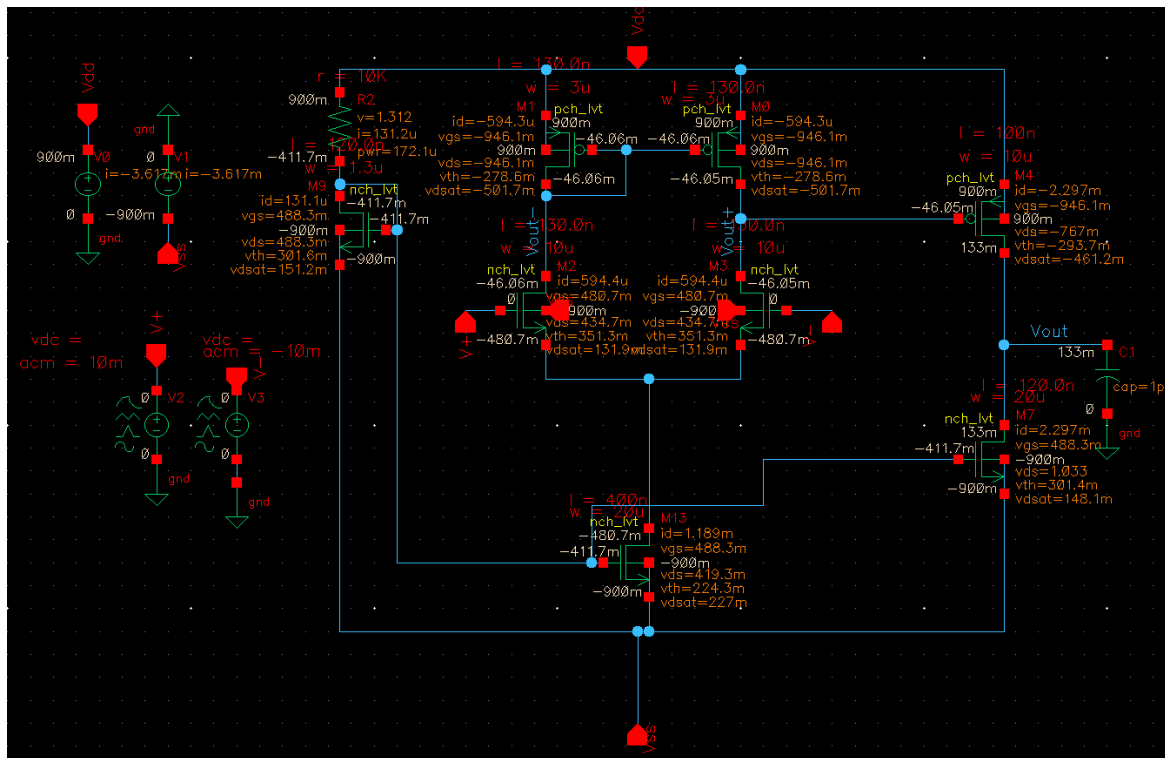


Figure 2: OTA Circuit Schematic Implemented in Cadence with sizing and DC operating points displayed

Simulation Results & Discussion

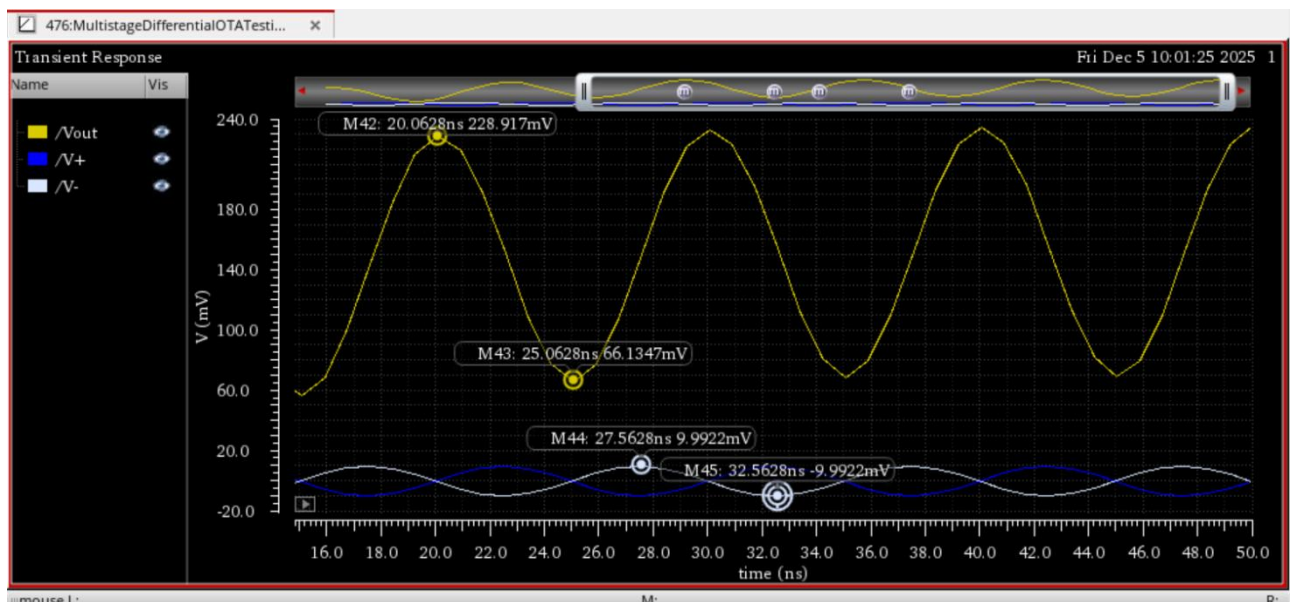


Figure 3: Transient Response of the Circuit

As seen above in Figure 3, the transient response clearly shows a large gain at the output with no distortion. We do notice a slight phase shift but that is not within the scope of our requirements.

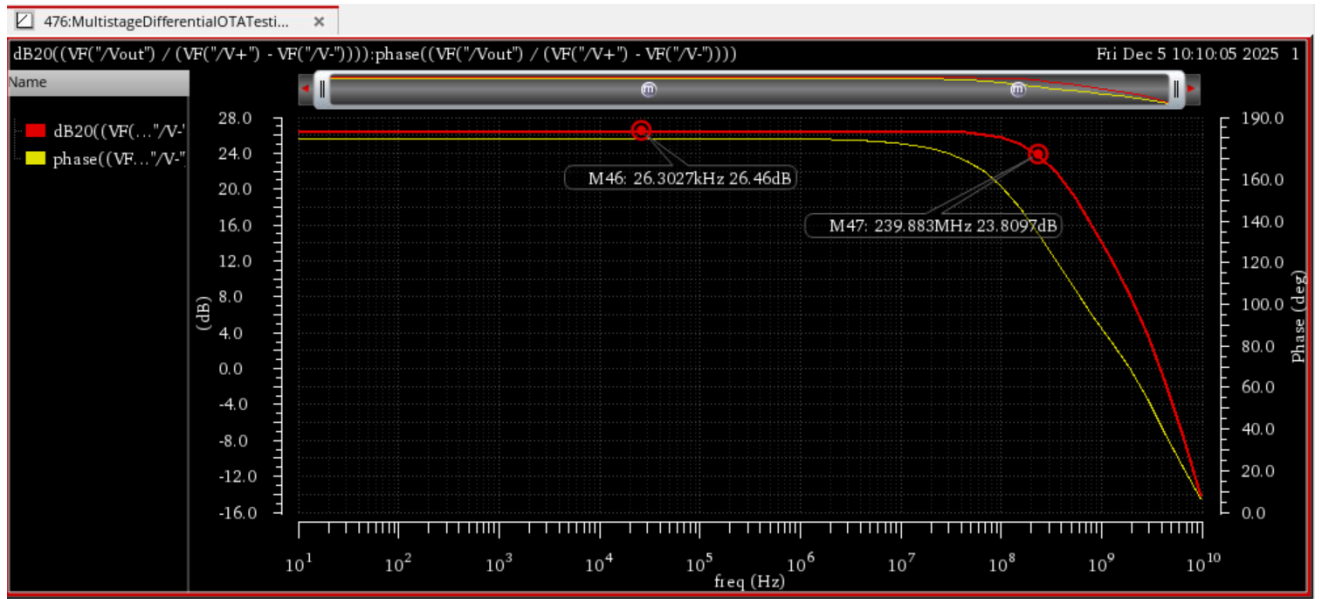


Figure 4: AC Response of the Circuit (Differential-Mode)

As seen above in Figure 4, the AC response in differential mode gives ~ 24.46 dB which is well above the 10dB requirement. This fulfills the gain requirement of our design.

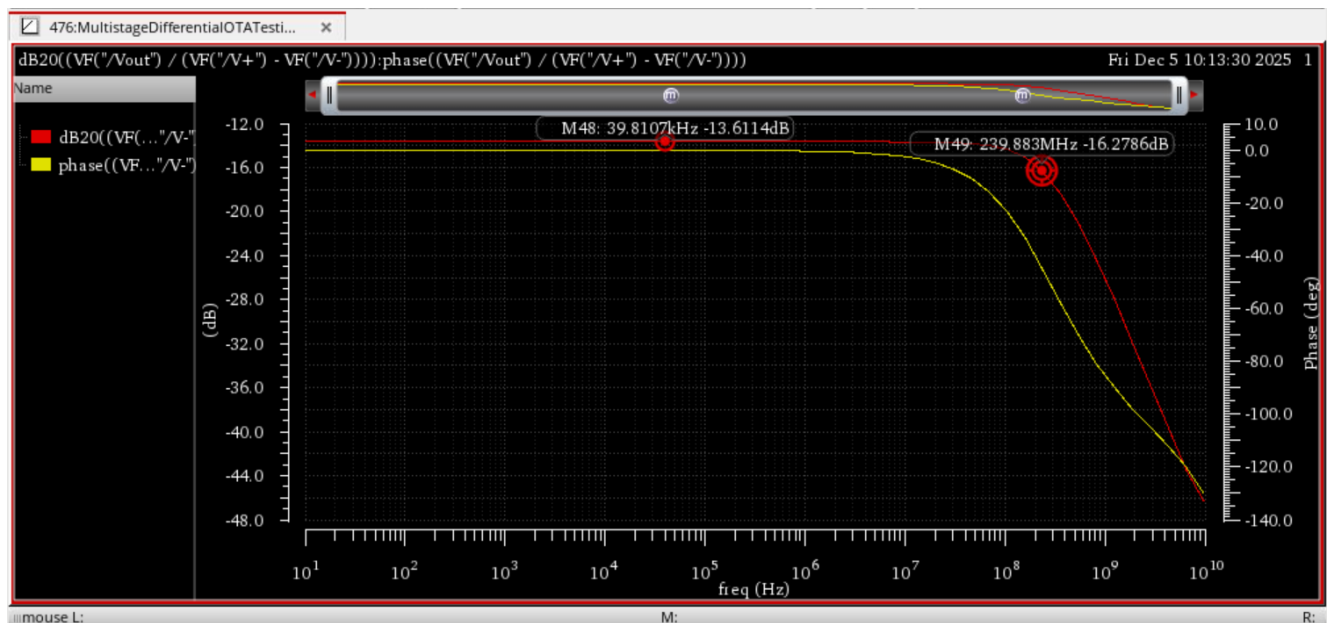


Figure 5: AC Response of the Circuit (Common-Mode)

As seen above in Figure 5, the AC response in common mode gives ~ -13.611 dB. This setup was done by using the same $V+$ to both of the input terminals of the differential pair. This was also used to calculate the CMRR of our design. This will be given below.

1	$20 \log(21.05)$	= 26.4650420034
2	$20 \log(.2085)$	= -13.6178788138
3	$\frac{21.05}{.2085}$	= 100.959232614

Figure 6: Calculations for CMRR

As seen above in Figure 6, we meet the design requirement of 100 CMRR. This was done using the values measured from Figures 4 and 5. We calculated 100.96 CMRR.

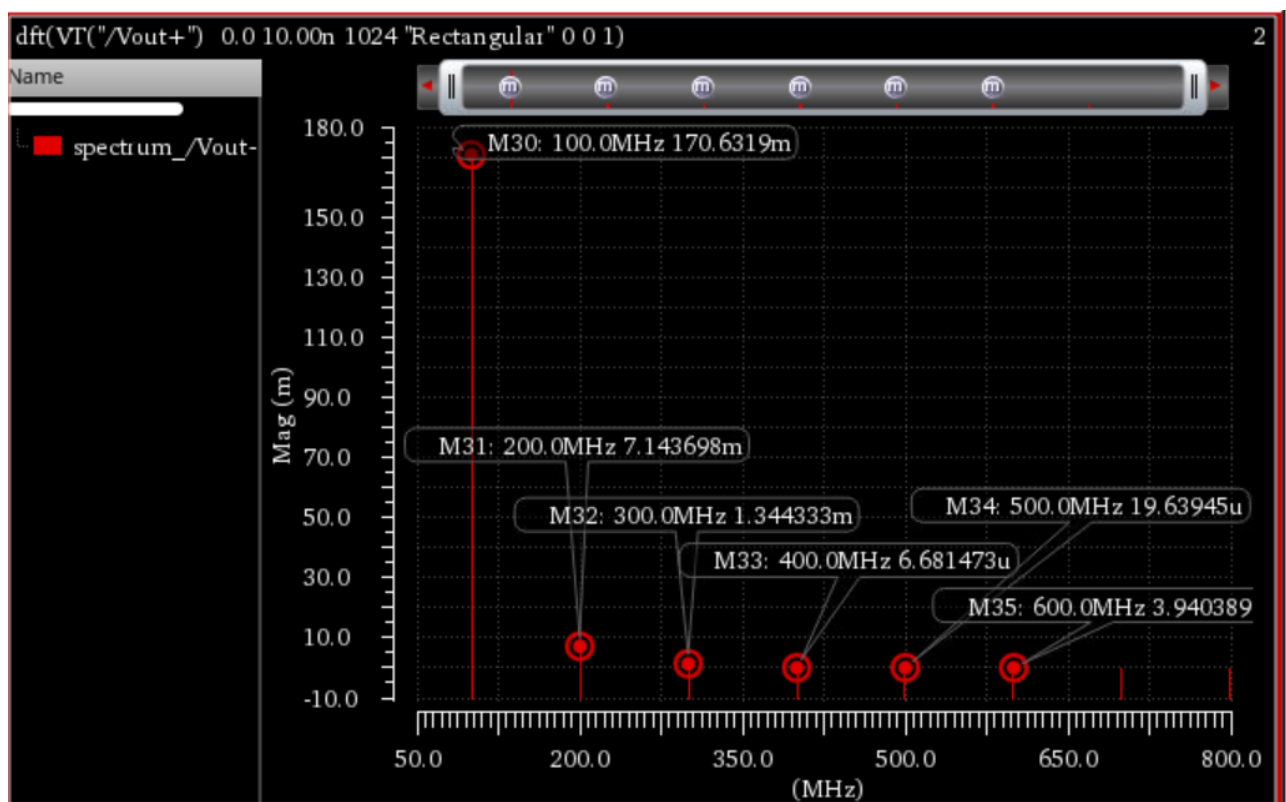
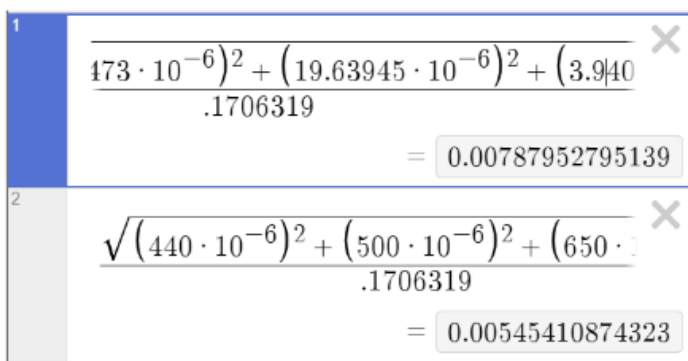


Figure 7: FFT of the Output Transient Signal at 100MHz

As seen in Figure 7, we observe a very large dominant frequency harmonic at our input harmonic and very low magnitude from the resultant harmonics. The points above were used to calculate the % Linearity and the calculations will be given below.

$$\text{THD}_F = \frac{\sqrt{V_2^2 + V_3^2 + V_4^2 + \dots}}{V_1},$$


1	$\frac{(173 \cdot 10^{-6})^2 + (19.63945 \cdot 10^{-6})^2 + (3.940)^2}{1706319}$	= 0.00787952795139
2	$\frac{(440 \cdot 10^{-6})^2 + (500 \cdot 10^{-6})^2 + (650 \cdot 10^{-6})^2}{1706319}$	= 0.00545410874323

Figure 8: Calculations of % Linearity

As seen in Figure 8, we calculated the % Linearity of our output to be 0.07%, which is less than the 1% design requirement. This meets specification. We also tested at 150MHz and got an even better result with 0.05% Linearity.

Conclusions

Overall, the circuit design functioned as intended and the output fulfilled the requirements of this design. We wanted to design for a Voltage gain: >10 dB and we measured ~24dB. We used a Load capacitor: 1pF as required. We wanted to design with a minimum of a -3 dB frequency: >200MHz and we measured ~240MHz. Our allowed Supply Voltage was +- 0.9 V. We wanted to design for a CMRR: >100 and we calculated 100.96 from simulation measurements. Finally we wanted to achieve 1% Linearity and we calculated it to be 0.07% which is well within the specification. Jun and I are satisfied with our results and think that our design holds up really well to what was asked.

References

[1] Dan Berco, EE 476 lecture8_OTA.pdf

[2] M. A. Mohammed and G. W. Roberts, "Scalable Multi-Stage CMOS OTAs With a Wide CL-Drivability Range Using Low-Frequency Zeros," in *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 70, no. 1, pp. 74-87, Jan. 2023, doi: 10.1109/TCSI.2022.3216201.



Multistage Differential Amplifier

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EE476



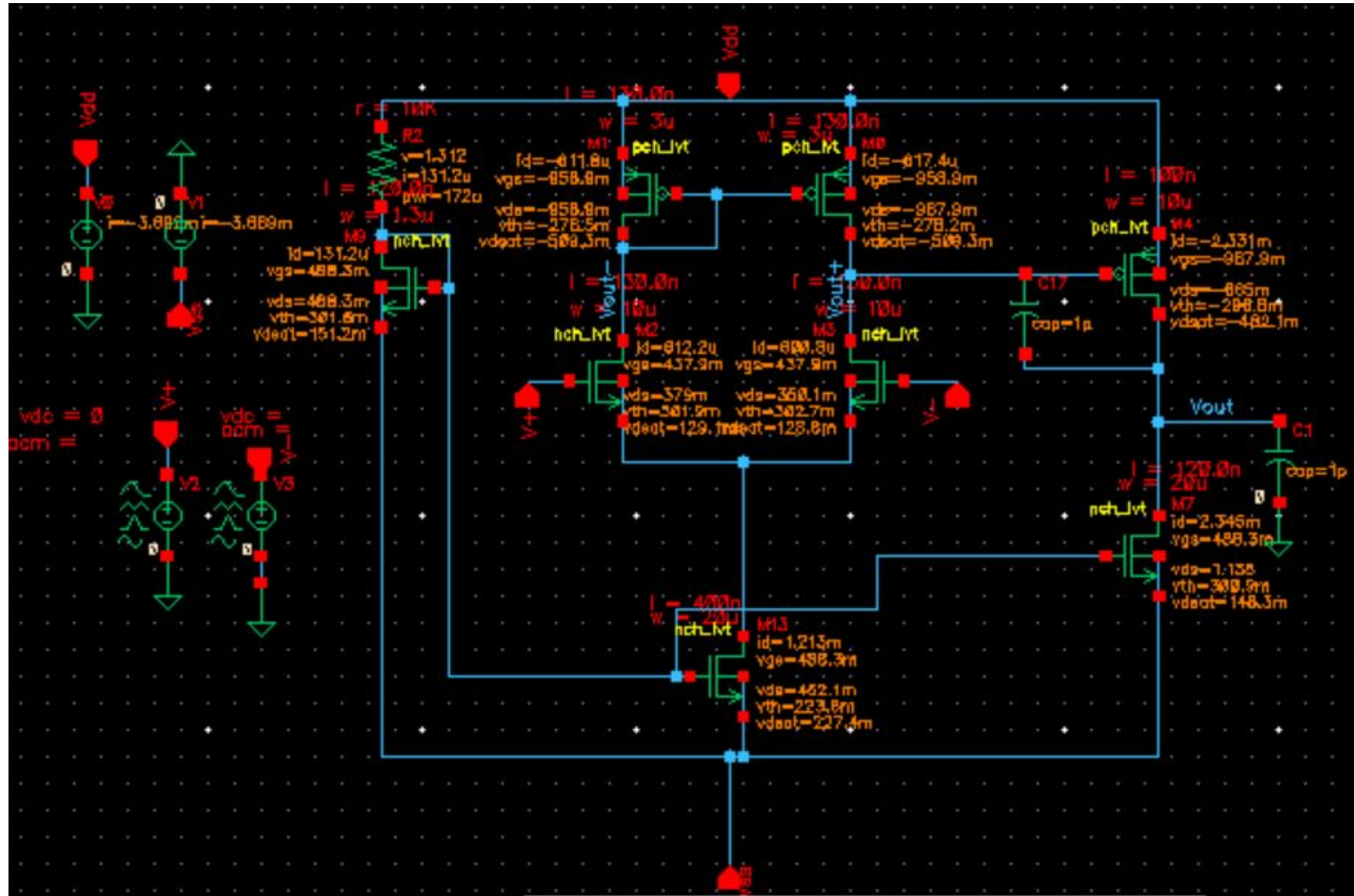


Design Specifications

- Voltage gain: >10 dB
- Load capacitor: 1pF
- -3 dB frequency: >200 MHz
- Supply Voltage: ± 0.9 V
- CMRR: >100
- 1% Linearity

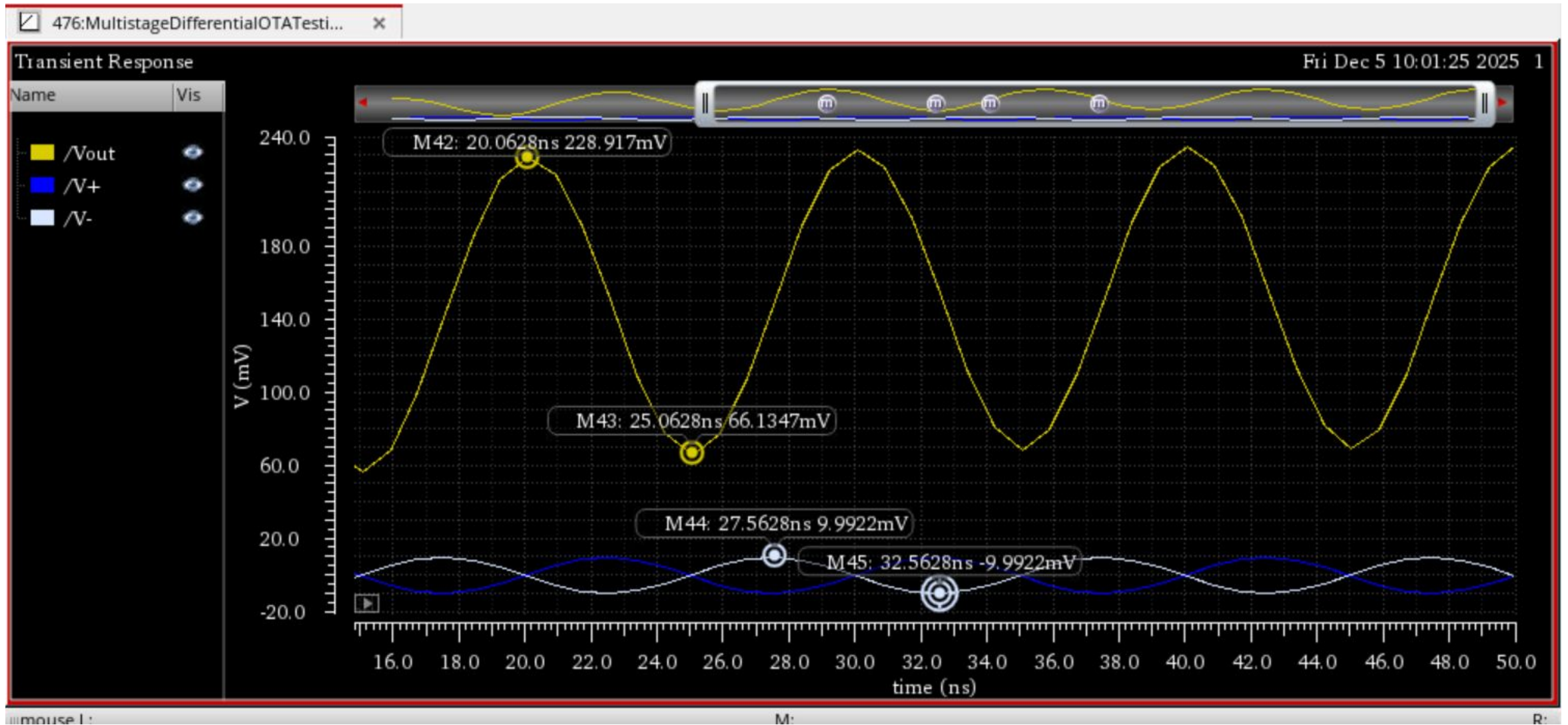


Design Schematic (Cadence Virtuoso)



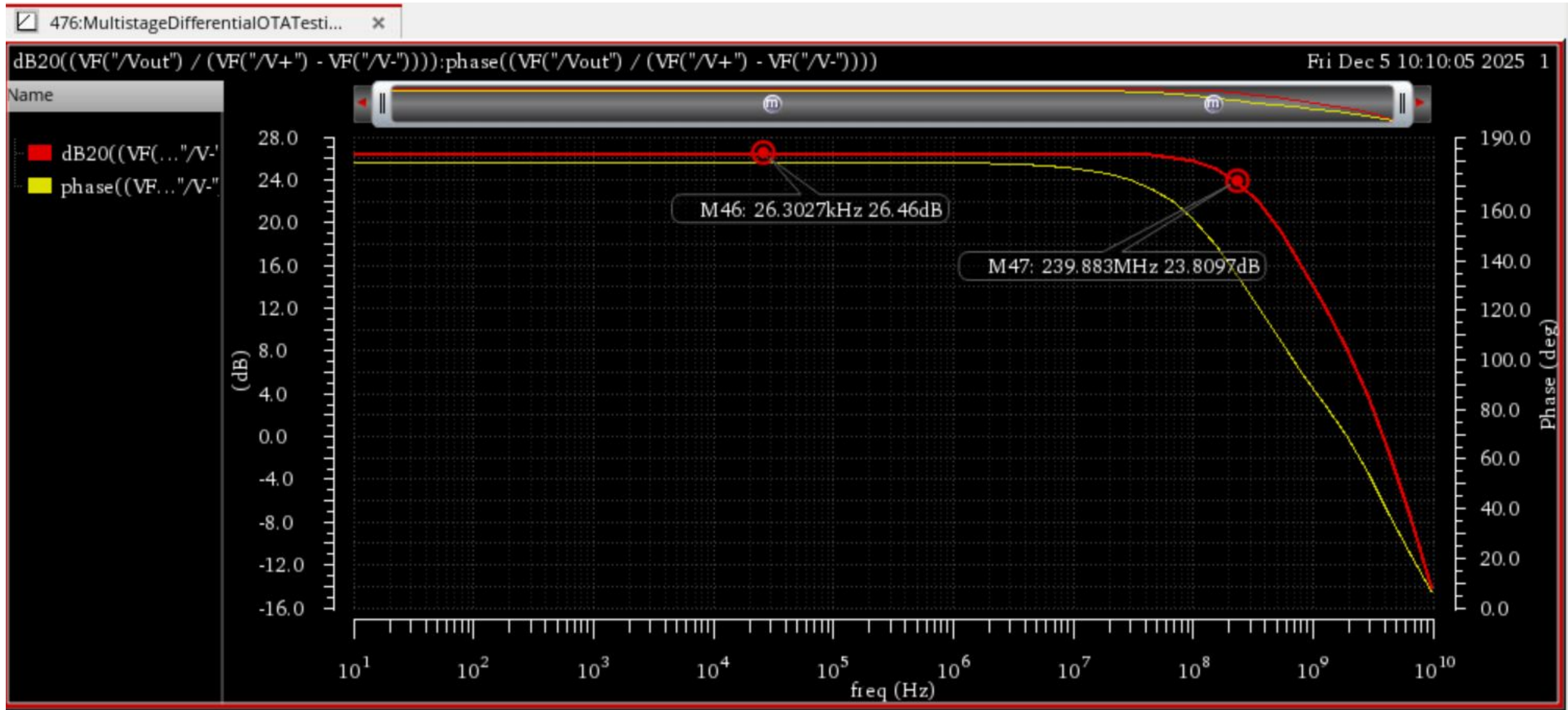


Transient Response



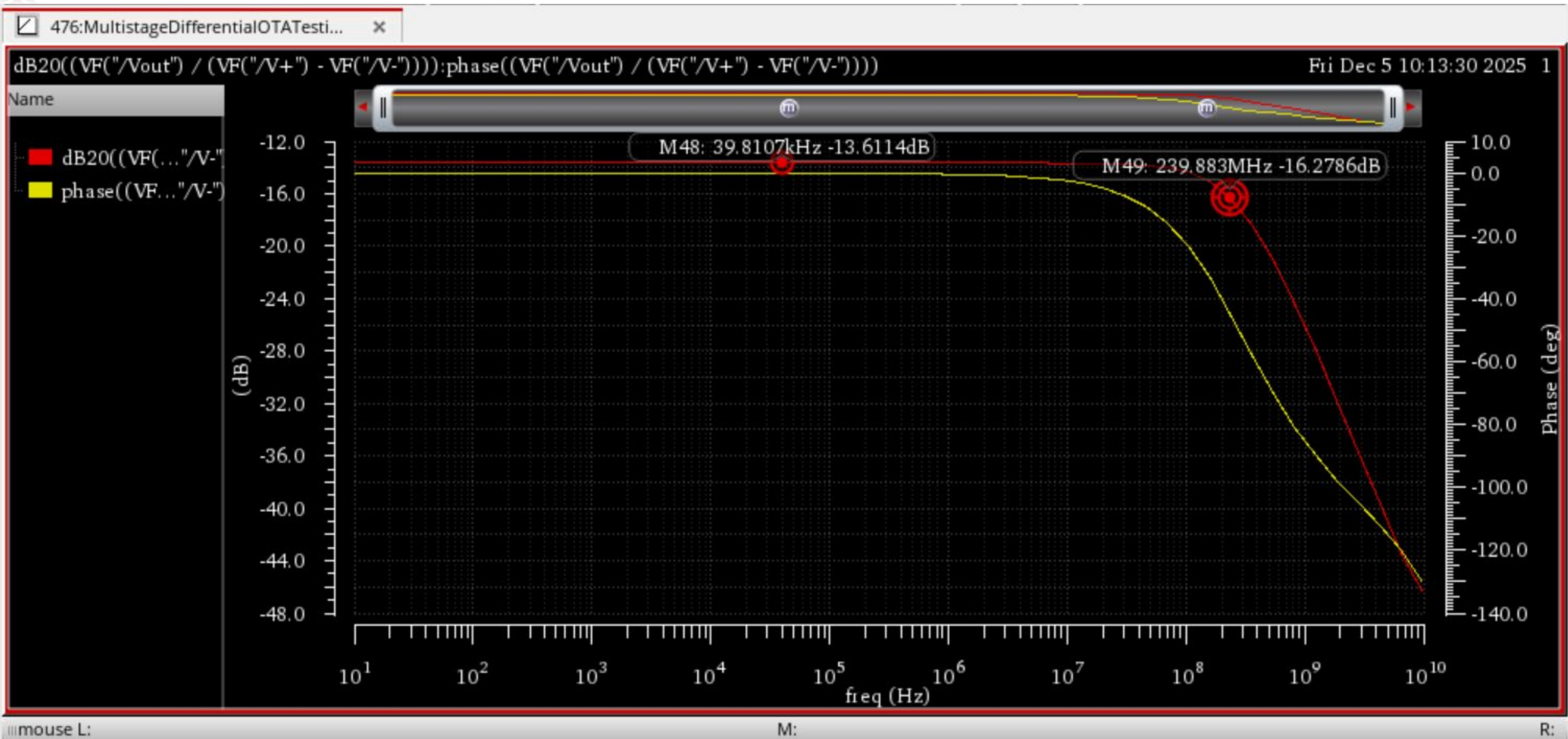


AC Response (dB) Differential-Mode





AC Response (dB) Common-Mode



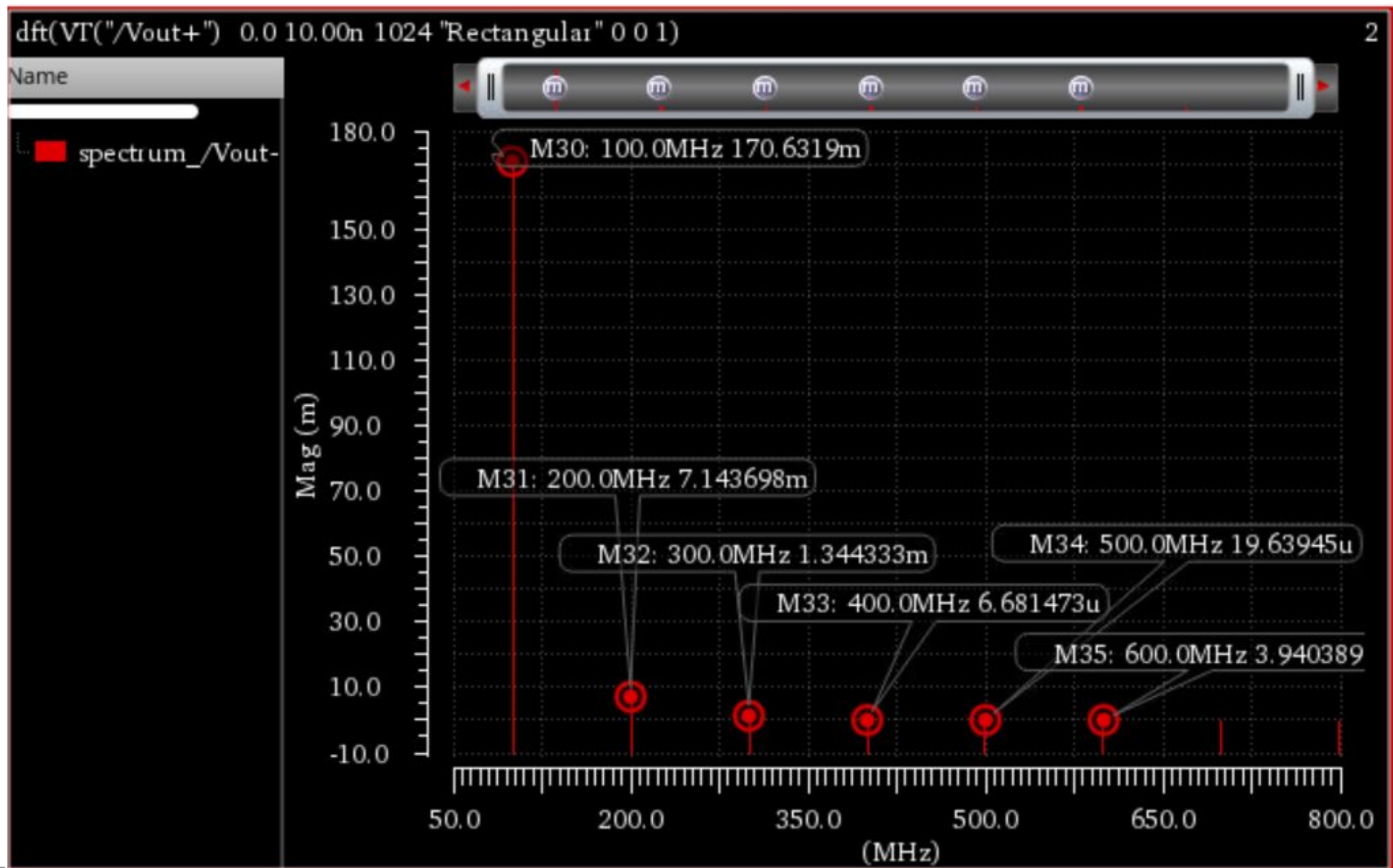


CMRR

1	$20 \log(21.05)$ $= 26.4650420034$ Differential	×
2	$20 \log(.2085)$ $= -13.6178788138$ Common Mode	×
3	$\frac{21.05}{.2085} = \frac{\text{Differential}}{\text{Common Mode}}$ $= 100.959232614$	×



FFT (100MHz)





Percent Linearity

$$\text{THD}_F = \frac{\sqrt{V_2^2 + V_3^2 + V_4^2 + \dots}}{V_1},$$

1	$\frac{(473 \cdot 10^{-6})^2 + (19.63945 \cdot 10^{-6})^2 + (3.940 \cdot 10^{-6})^2}{.1706319}$ $= 0.00787952795139$
2	$\frac{\sqrt{(440 \cdot 10^{-6})^2 + (500 \cdot 10^{-6})^2 + (650 \cdot 10^{-6})^2}}{.1706319}$ $= 0.00545410874323$



Design Results

- Voltage gain: 10 dB or higher
- Load capacitor: 1pF
- -3 dB frequency: greater than 200 MHz
- Supply Voltage: ± 0.9 V
- CMRR: 100.9
- 0.7% Linearity